

Aayush Gnawali

DevOps • DevSecOps • AI/ML • Backend

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PORTFOLIO

Summary

Final-year B.E. Information Technology student with hands-on experience building backend systems, containerized infrastructure, and applied AI/ML projects. Comfortable across the stack that keeps services running: Linux fundamentals, Docker, CI/CD concepts, and FastAPI-based APIs, with a growing focus on DevSecOps and cloud deployment. Seeking remote opportunities to work across backend, DevOps, and applied AI on systems other engineers can depend on.

Experience

DevOps Intern

2026

Lego Tech Pvt. Ltd.

- Applied Linux fundamentals and Bash scripting to support day-to-day system and development workflows.
- Built and ran containerized services using Docker and Docker Compose, working with YAML-based configuration.
- Gained practical exposure to AWS core services, including EC2, for deploying and managing infrastructure.
- Learned and applied CI/CD concepts and server fundamentals alongside a professional DevOps team.
- Used Git for version control across development and infrastructure tasks.

Backend & QA Intern

2026

E-Digital Nepal

- Developed a production module using FastAPI as part of a live ERP system.
- Contributed to backend API development alongside the production development team.
- Performed QA testing to validate functionality and support reliable releases.
- Collaborated with developers during implementation and testing cycles to identify and resolve issues.

Projects

Rule-Based Self-Healing Web Automation System

Python, Automation, UI Monitoring, Telemetry

- Built a Python-based system that detects UI drift, tracks browser state, and recovers failed workflows automatically using rule-based healing logic instead of manual intervention.
- Designed resilient automation to handle shifting selectors and UI structure during live application updates.

HippyHub E-Commerce Website

[GitHub](#)

React, FastAPI, PostgreSQL, Docker, nginx

- Built a 3-tier e-commerce platform for hemp products and Dhaka-pattern clothing with a React frontend, FastAPI backend, and PostgreSQL database.
- Containerized and ran the full stack with Docker as part of DevOps training.

House Price Prediction Deployment

[Live Demo](#)

Python, Scikit-learn, Streamlit, Pandas

- Built and deployed a regression model with a real-time inference pipeline and performance-monitoring visualizations.
- Solved feature engineering issues where early versions overfit to training-data patterns that did not generalize.

IoT-Based Smart Home Monitoring System

ESP8266, HTTP, MySQL, Arduino IDE

- Built an ESP8266 microcontroller-based system for remote monitoring of home environment parameters using HTTP for real-time data transmission and MySQL for storage.

3D Pomodoro Website

[Live Demo](#)

JavaScript, Three.js, GitHub Pages

- Built an interactive 3D productivity timer with mesh rendering and rotational mechanics driven by timer state.
- Tuned real-time 3D rendering performance in-browser while timer state updated continuously.

Digital Desktop Watch

[GitHub](#)

Python, PyQt6

- Built a desktop clock application with real-time updates, custom widgets, and alarm functionality using PyQt6.

Skills

Backend: FastAPI, REST APIs, Async SQLAlchemy, Pydantic, JWT Authentication, OOP

DevOps: Docker, Docker Compose, Linux, Bash, YAML, CI/CD Concepts, Git, GitHub

AI / ML: PyTorch, Scikit-learn, Pandas, NumPy, Model Deployment
Cloud & Deployment: Render, Streamlit Cloud, GitHub Pages, AWS, Deployment Fundamentals
Databases: PostgreSQL, MySQL, SQL
Frontend: React, Next.js, TypeScript, Tailwind CSS
Tools: Git, GitHub, Selenium, Streamlit, Virtual Environments, Figma
Other: Data Structures & Algorithms, Technical Writing, Problem Solving

Education

B.E. in Information Technology	2022 – Present
<i>Everest Engineering College (Pokhara University), Lalitpur, Nepal</i>	
Higher Secondary Education (+2 Science)	2020 – 2022
<i>Trinity International College, Kathmandu, Nepal</i>	

Certifications

AI/ML Workshop	2025
<i>ITSNP</i>	
Intensive training on neural network architectures and implementation, including CNNs.	
DevOps and Cloud Computing Training	2026
<i>Upskill Nepal</i>	
Hands-on cloud and DevOps training covering deployment, automation, and infrastructure best practices.	
Robotics Training	2025
<i>Everest Engineering College</i>	
Embedded systems and automation fundamentals.	

Achievements & Leadership

Best Minor Project	2025
<i>Everest Engineering College</i>	
Won best minor project among all 6th semester submissions for technical execution, problem solving, and presentation.	
Chief Content Writer, Hult Prize Organizing Committee	2025
<i>Everest Engineering College</i>	
Led content strategy and communications; team ranked in the Top 25% in South Asia among roughly 300 organizing committees.	